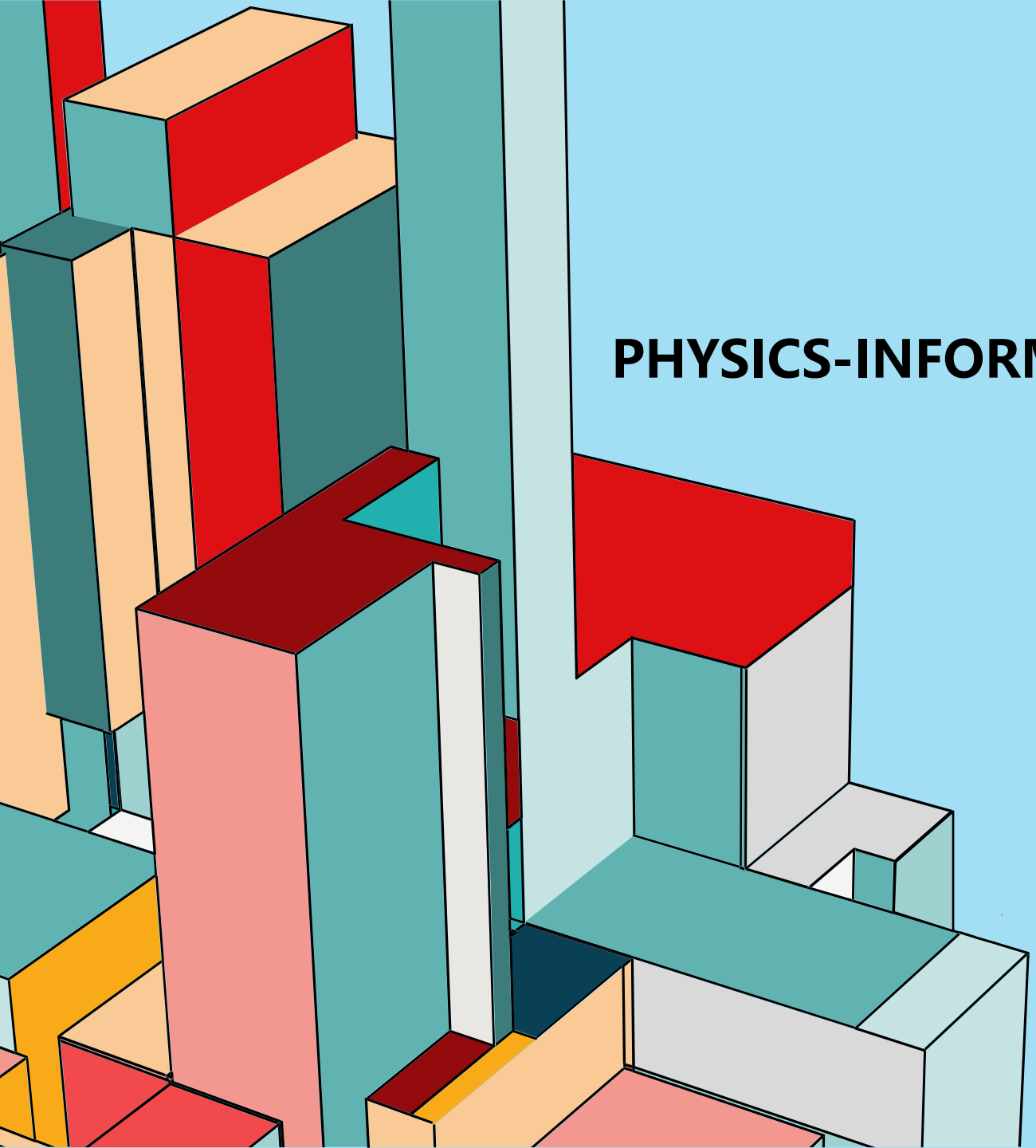


PHYSICS-INFORMED NEURAL NETWORK (PINN)

BASED ON:
A PINN FRAMEWORK FOR GENERIC BIOREACTOR MODELLING

PRE BY :
WANG ŻEZHONG & ZHAO YUXI

BACK GROUND DEFINITION

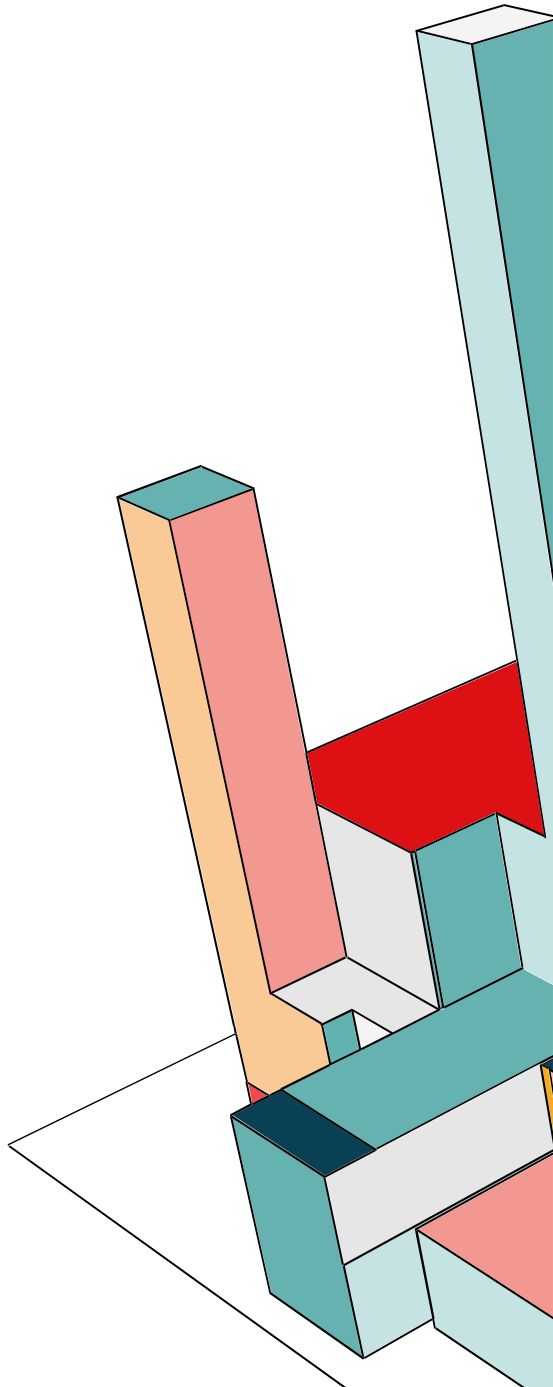
Stirred-Tank Fed-Batch Bioreactor

A generic perfectly mixed fed-batch bioreactor is described by mass balances:

- $\frac{dC}{dt} = Sr(C,u) - DC + DC_{in}$ (1)
- $\frac{dV}{dt} = F$

Where:

- $C \in \mathbb{R}^n$:state vector (biomass, substrate, products, etc.)
- $r(C,u)$:reaction kinetics
- S: stoichiometric matrix
- F : Feed rate
- $D = \frac{F}{V}$:dilution rate (2)
- $u(t)$:control inputs (feed rate, temperature, etc.)
- V: reactor volume



TRADITIONAL APPROACHES & LIMITATIONS

CLASSICAL MECHANISTIC MODELING

The main idea

- Based on ODEs:

$$\frac{dX}{dt} = f(X, t, \theta)$$

Advantages:

- Interpretable parameters
- Physically consistent
- Low data requirement

Limitations:

- Model mismatch
- Difficult parameter estimation
- Sensitive to noise

PURE DATA-DRIVEN NEURAL NETWORKS

The main idea

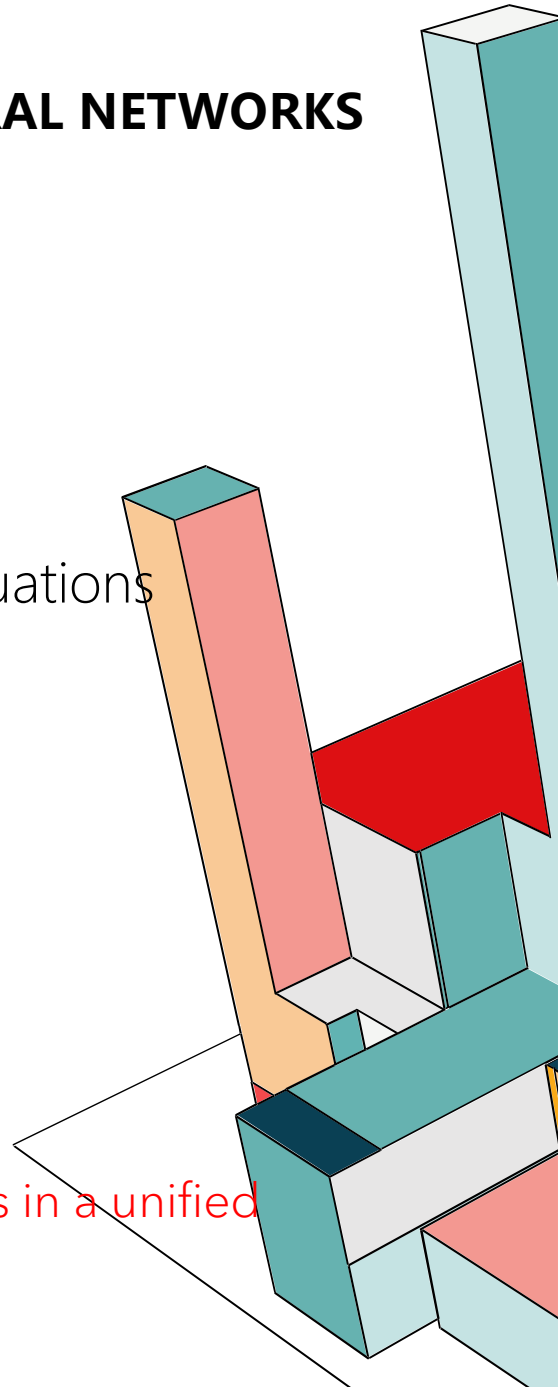
$X(t)=N\theta(t)$

Advantages:

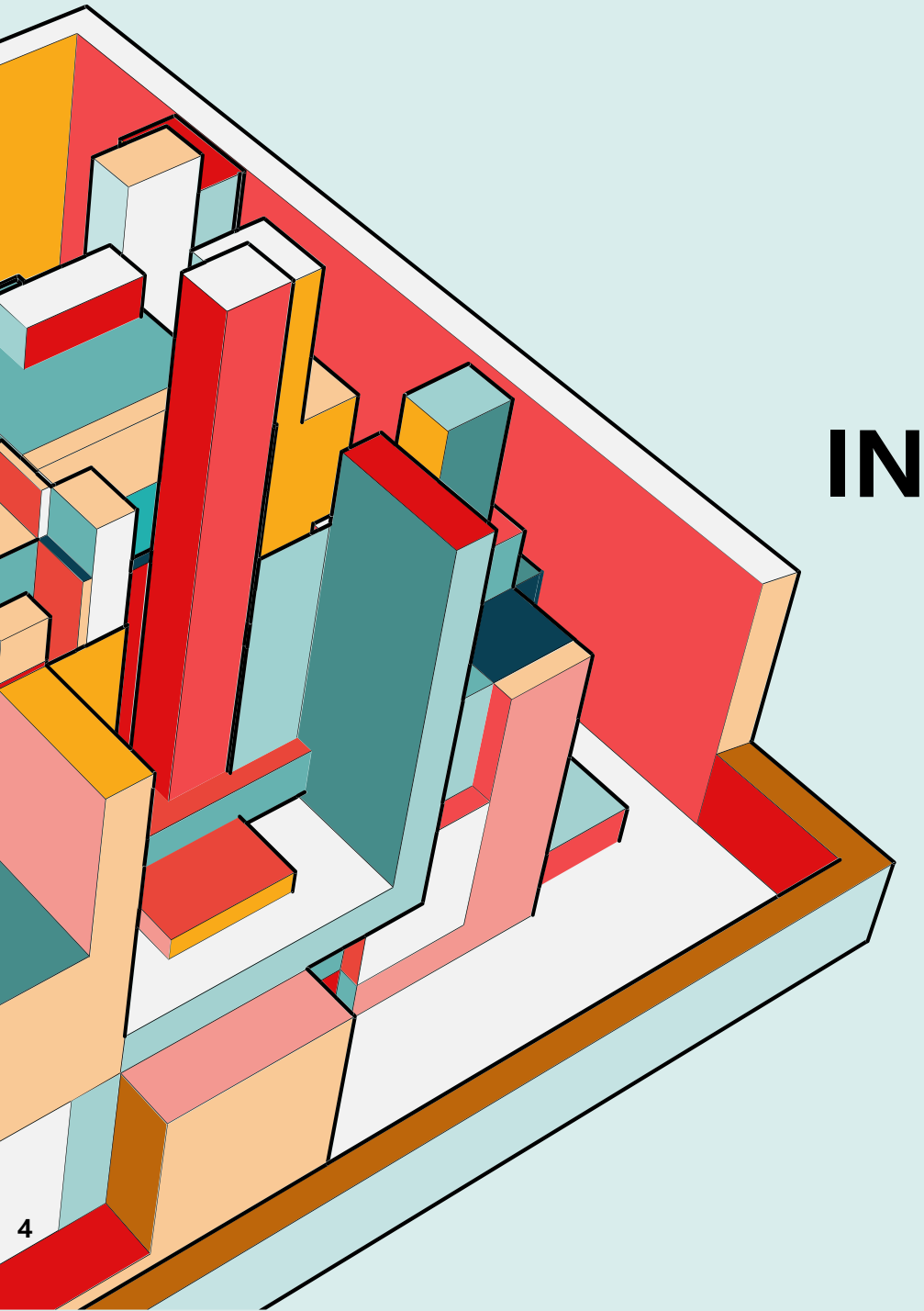
- Strong fitting capability
- No need for explicit equations

Limitations:

- No physical guarantee
- Poor extrapolation
- Large data requirement



Can we combine physical laws and neural networks in a unified framework?



INTRODUCING PINNS

MAIN IDEA

A Physics-Informed Neural Network (PINN) combines:

- Data fitting
- Physical laws (ODEs or PDEs)

Standard Neural Network

- $X_\theta(t)$

Loss:

- $\mathcal{L}_{data} = \sum_i |X_\theta(t_i) - X_i|^2$

PINN

Neural network still predicts:

- $X_\theta(t)$

But we enforce physics:

- $\frac{dX_\theta}{dt} = f(X_\theta, t, \theta)$

Physics residual:

- $r(t) = \frac{dX_\theta}{dt} - f(X_\theta, t, \theta)$

Total Loss

- $\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics}$

PINN FORMULATION FOR FED-BATCH SYSTEM

Governing Equations (Logistic Fed-Batch)

- x *: Biomass concentration*
- $\mu = \mu_{\max}\left(1 - \frac{X}{X_{\max}}\right)$
- $D(t) = \frac{F(t)}{V(t)}$
- $\frac{dX}{dt} = \mu X - DX \quad \rightarrow \quad X(t)$
- $\frac{dV}{dt} = F(t) \quad \rightarrow \quad V(t)$

Neural Network Representation

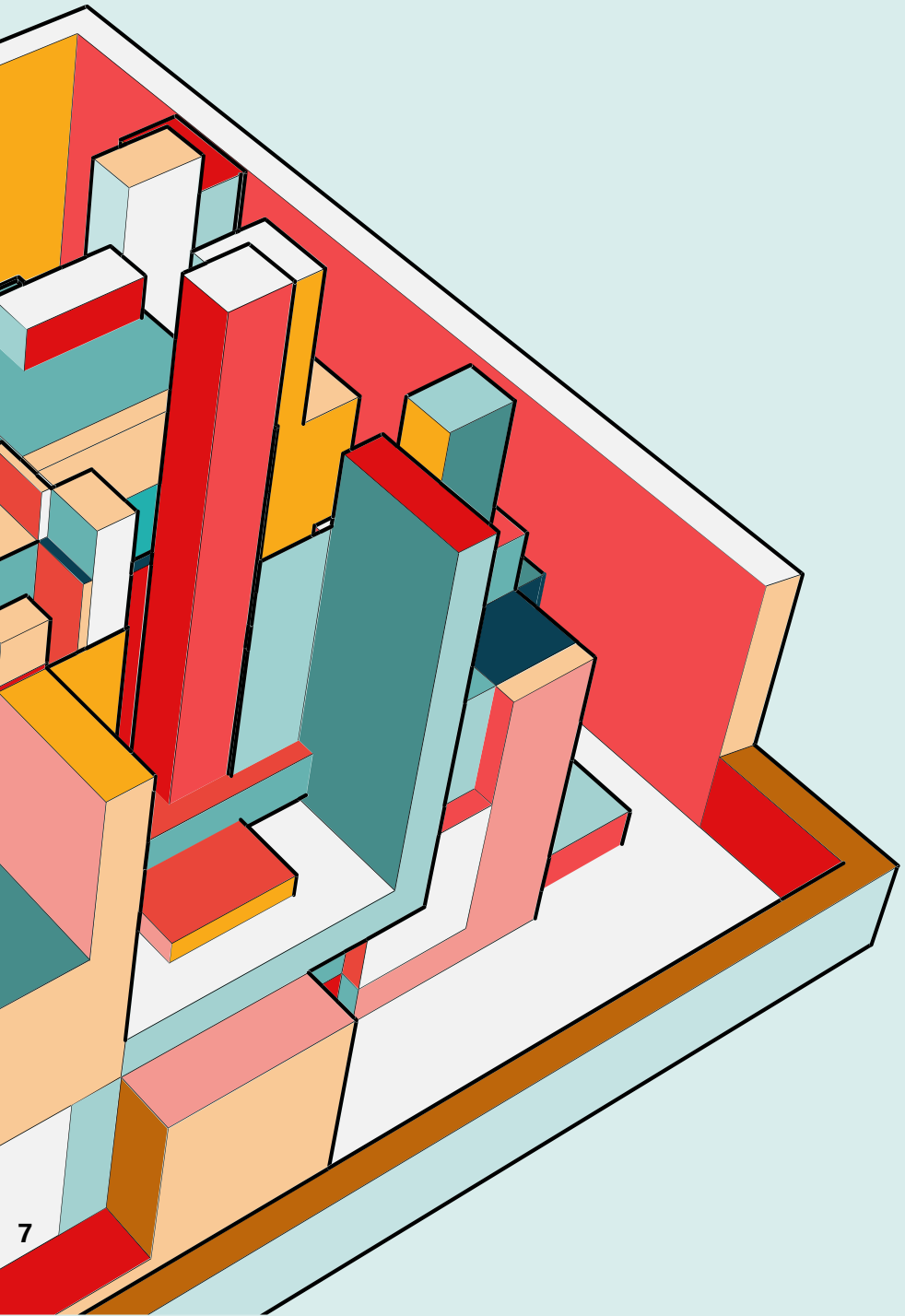
- $t \rightarrow NN \rightarrow (X_\theta, (t)V_\theta(t))$

Physics Residuals

- $r_X(t) = \frac{dX_\theta}{dt} - (\mu_\theta X_\theta - DX_\theta)$
- $r_V(t) = \frac{dV_\theta}{dt} - F(t)$

Total Loss

- $\mathcal{L} = \mathcal{L}_{data} + \lambda_1 \sum | r_X |^2 + \lambda_2 \sum | r_V |^2$



EXPERIMENTAL STUDY

EXPERIMENTAL STUDY

DATA SOURCE & MEASURED VARIABLES

Data are taken from data.csv[1] containing multi-batch fed-batch experiments.

Each sample includes time t hours, noisy measurements X,S,Pt,Pm,V and the input feed rate F.

X (g/L): Biomass concentration S(g/L): Substrate concentration

Pt (g/L) : Total product concentration

Pm(g/L) : Secreted product concentration

The file also provides corresponding noise free trajectories Xtrue,Strue,Pt,true,Pm,true used only for evaluation.

Fixed process constants used in the physics model: Sin=20g and yield Y=7.3

EXPERIMENTAL STUDY

BATCH LEVEL PARTITION

- We use a batch-level split to avoid leakage between train and test trajectories.
- Test set: Batch 16 (annotated in code as an optimal control profile, most meaningful for extrapolation).
- Training set: Batches 1–15.
- No explicit validation set in this version; generalization is promoted via stochastic batch sampling during training.

EXPERIMENTAL STUDY

INITIAL CONDITIONS AS INPUTS

- Initial conditions are extracted from the first point of each batch:

$$(X0, S0, Pt0, Pm0)$$

  and replicated across all samples of that batch.

- The network learns a conditional mapping:

$$(X, S, Pt, Pm) = f\theta(\tau, V, F, X0, S0, Pt0, Pm0)$$

enabling one model to handle multiple batches with different initial states and inputs.

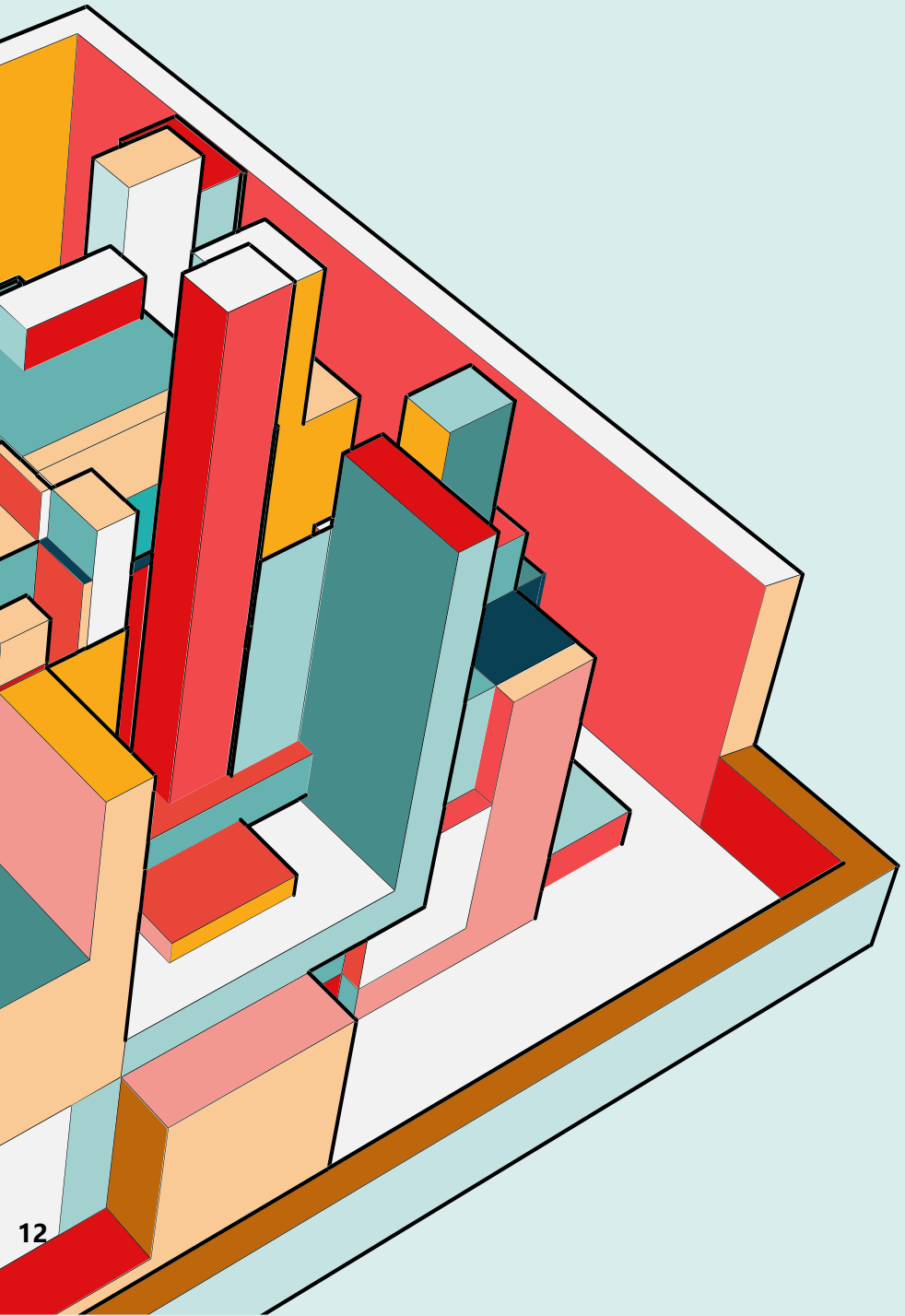
- Inputs (7D): $u = [\tau, Vexact, F, X0, S0, Pt0, Pm0]$
- Outputs (4D): $y = [X, S, Pt, Pm]$
- Architecture: 4 hidden layers, 128 neurons each, tanh activation

EXPERIMENTAL STUDY

TRAINING PROTOCOL & MODEL SELECTION

- Training length: totalEpochs=50000.
- Fixed loss weighting: λ=0.5.
- Learning-rate schedule:10^{-4}10−3→5×10−4→10−4 (piecewise decay).
- Model selection: every 500 epochs, compute test R^2 on **Batch 16** for Pm; keep the best model.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2 + 10^{-10}}$$



EXPERIMENTAL RESULTS

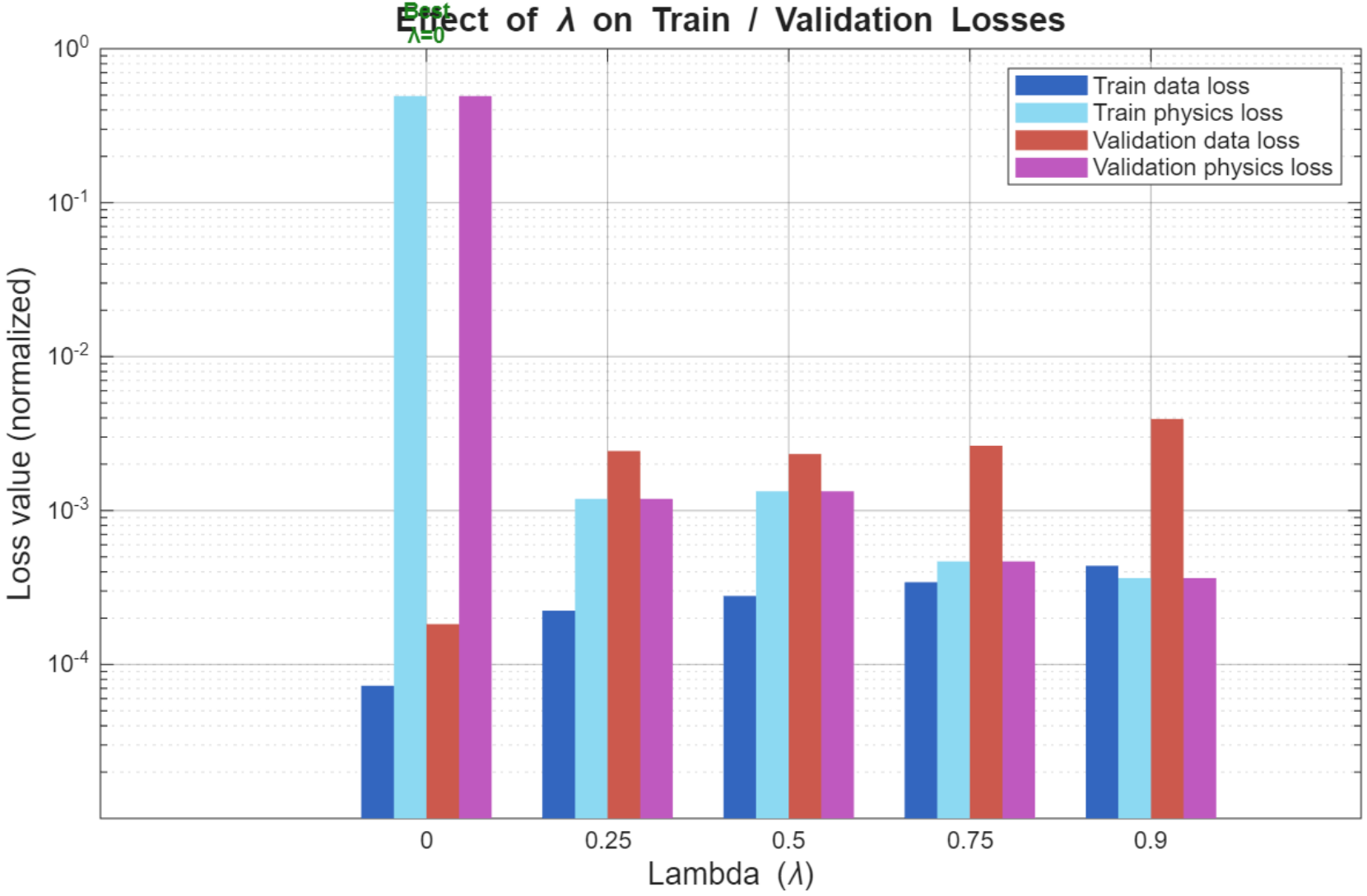
EXPERIMENTAL RESULTS

RESULTS OVERVIEW

- We report:
- (i) sensitivity to the loss weight λ
- (ii) training convergence
- (iii) accuracy metrics (R^2) on Train/Val/Test
- (iv) time-series performance on the unseen Test batch (Batch 16).
- The evaluation focuses on the four predicted states: X, S, Pt, Pm.
- Batch 16 is used as the main generalization benchmark (unseen input profile).

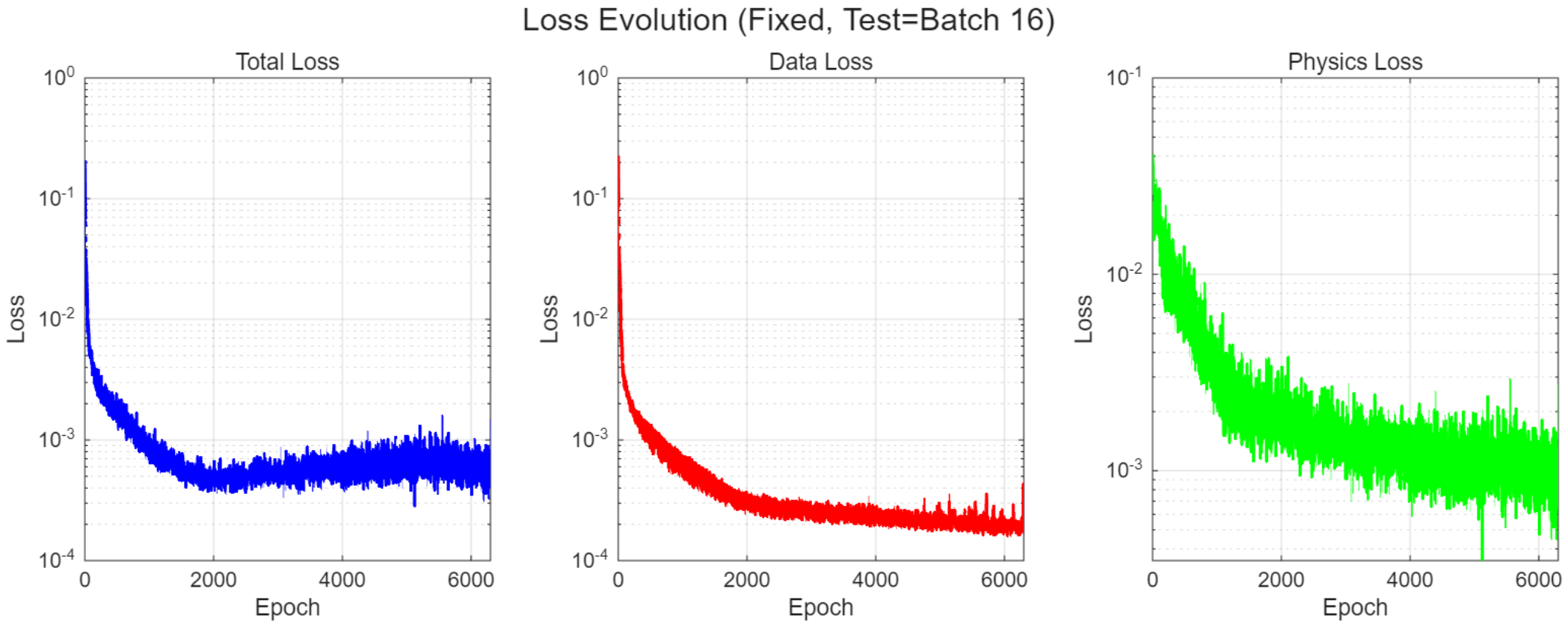
EXPERIMENTAL RESULTS

EFFECT OF λ (DATA VS PHYSICS TRADE-OFF)



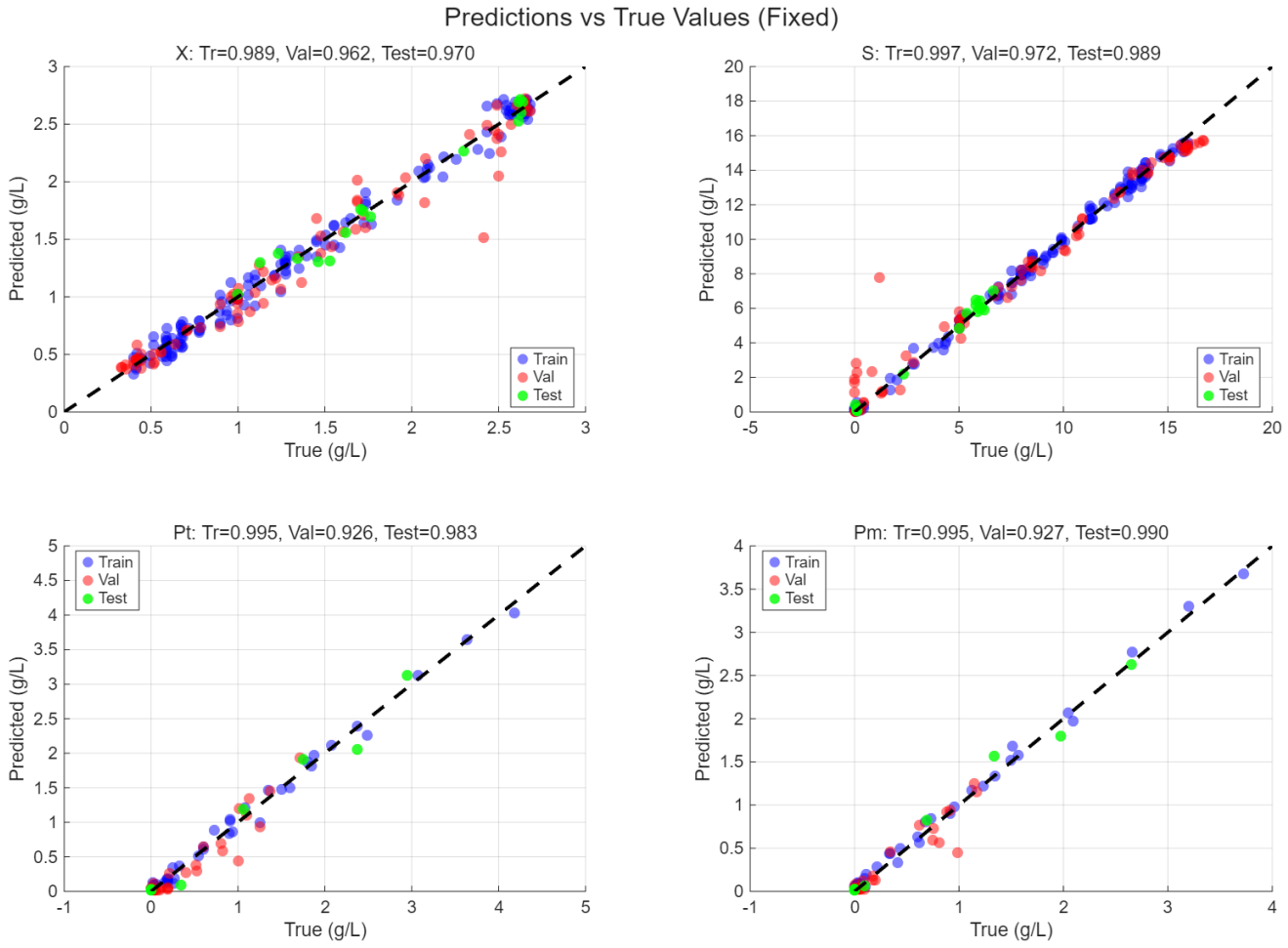
EXPERIMENTAL RESULTS

TRAINING CONVERGENCE



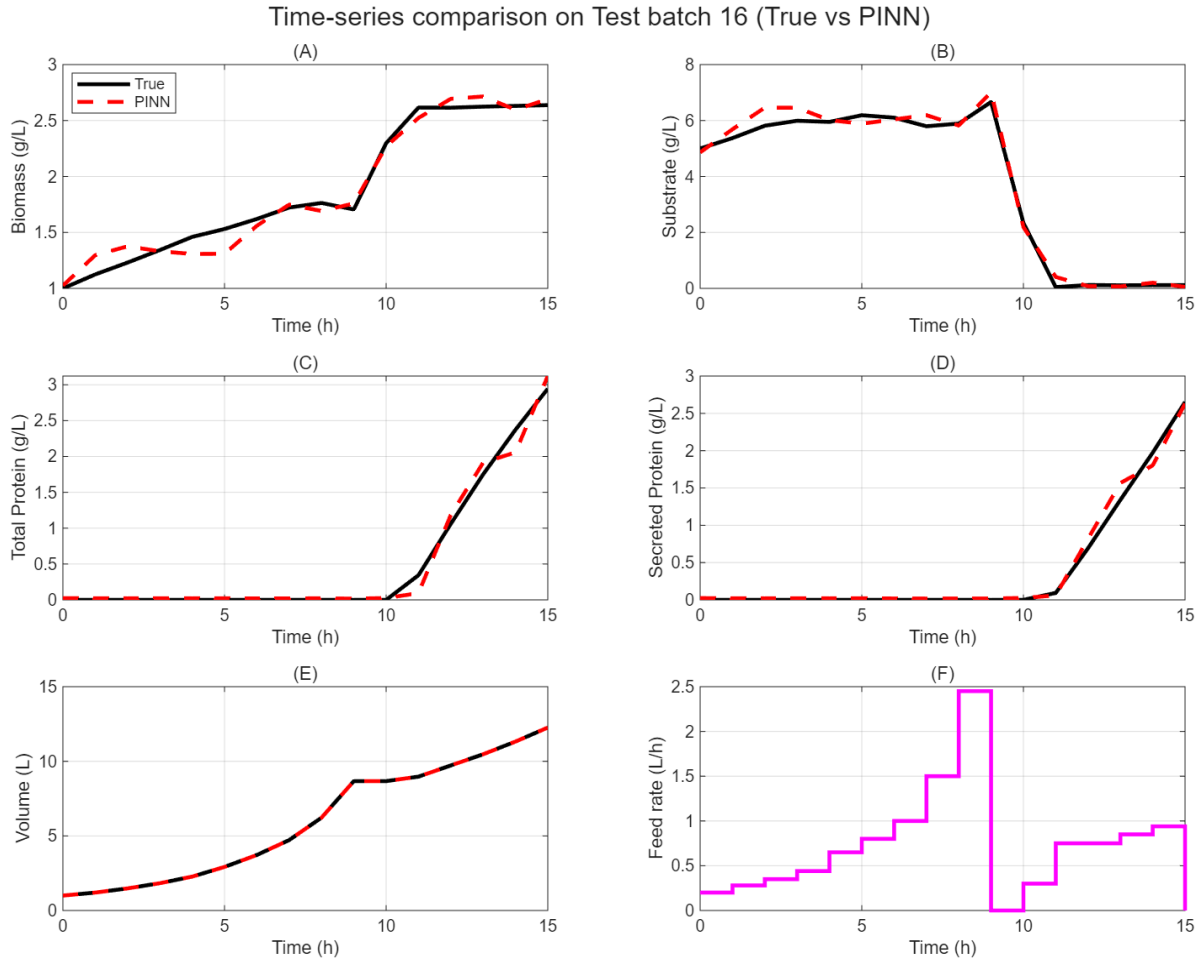
EXPERIMENTAL RESULTS

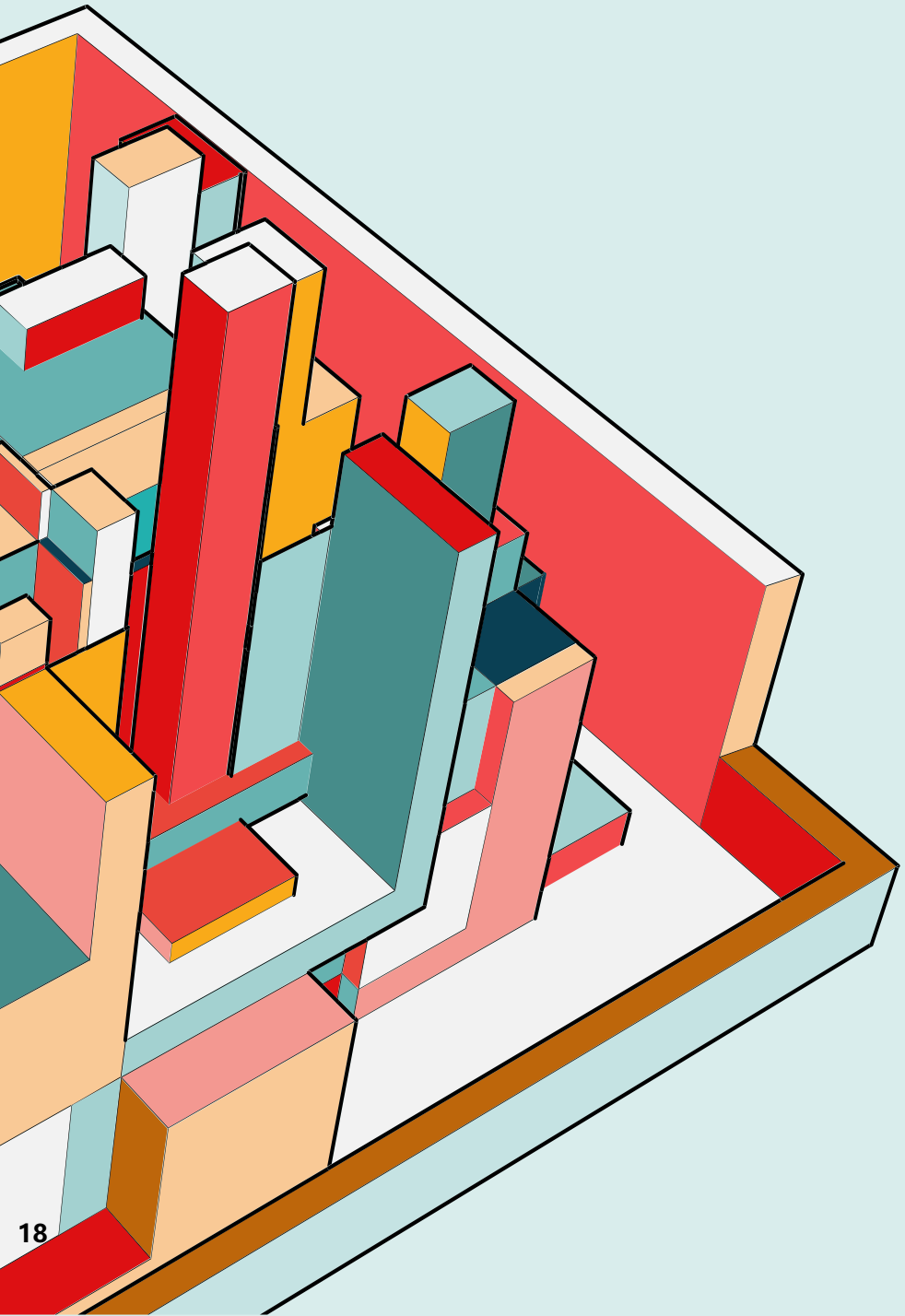
TRAINING CONVERGENCE



EXPERIMENTAL RESULTS

QUANTITATIVE ACCURACY





DISCUSSION & CONCLUSION

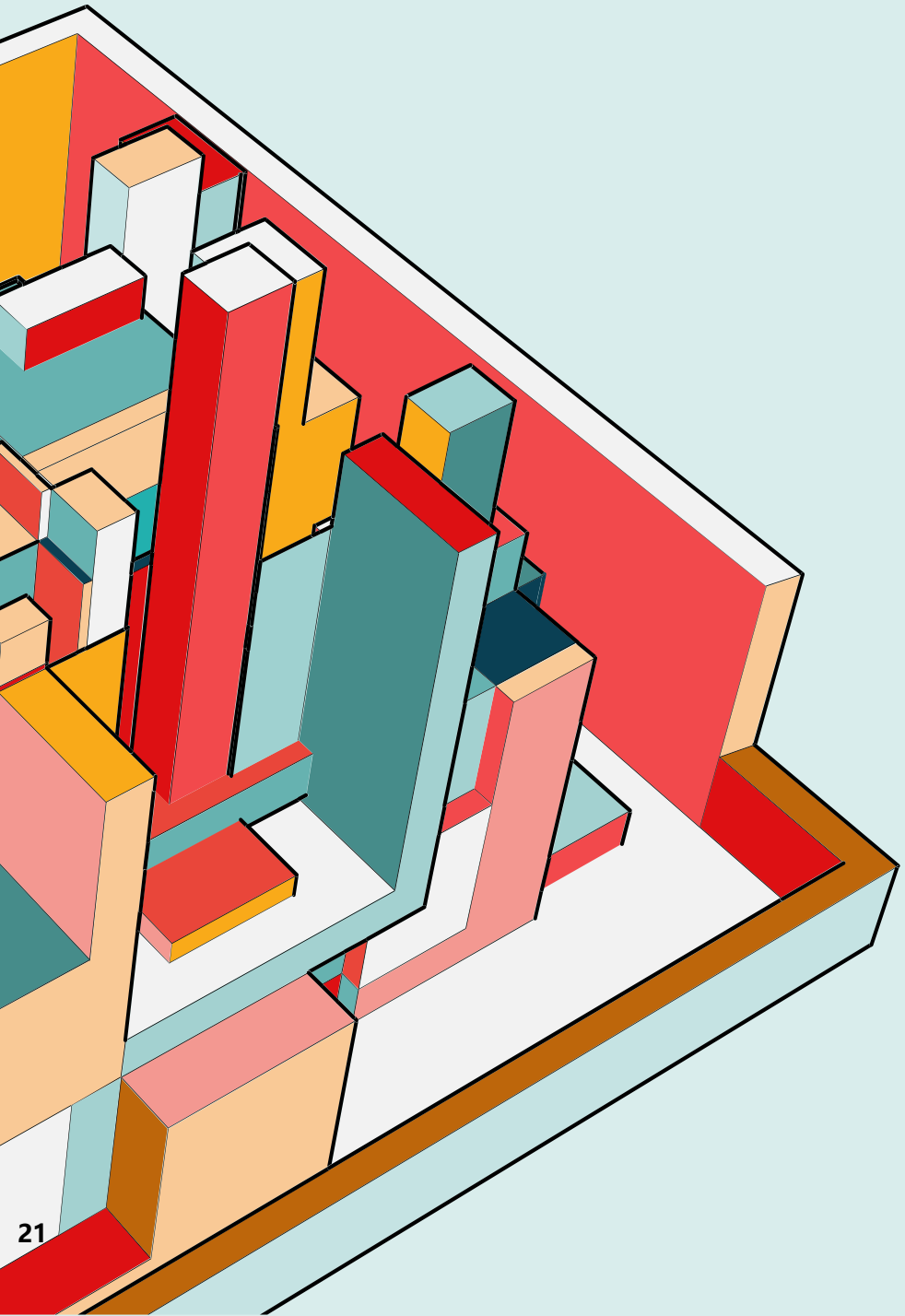
DISCUSSION & CONCLUSION

THREE MODELING OPTIONS

- Pure physics model: interpretable and physically consistent, but accuracy drops if kinetics/parameters are wrong.
- Pure NN (data-driven): best accuracy when data are sufficient, but can overfit and may violate physics.
- PINN: adds physics as a regularizer → slightly less accurate than pure NN in our runs, but more reliable with limited/noisy data and better physical plausibility.
- Takeaway: use pure NN when data are rich; use PINN when data are scarce or you need physically consistent predictions.

THANKS!





REFERENCES

REFERENCES

[1] Thirugnanasambandam, Monesh kumar et al. “A Physics-Informed Neural Network (PINN) Framework for Generic Bioreactor Modelling.” *Computers & chemical engineering* 203 (2025): 109354. Print.